F1 Sponsorship ROI Analytics

Post-2014 Hybrid Era — Growth, Consistency & Risk Intelligence for Corporate Sponsors
Newton School of Technology · Data Visualization & Analytics · Capstone 2

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Which F1 teams deserve your sponsorship budget?

01

The $2.6B market

F1 runs 24 races across six continents, delivering live broadcast exposure to affluent global audiences — a uniquely valuable sponsorship stage.

02

Gut-driven decisions

CMOs and investment managers allocate millions based on brand reputation or recent headlines — not forward-looking evidence.

03

The key question

Which constructors offer the highest Growth ROI and Consistency in the post-2014 Modern Hybrid Era?

04

The solution

A Python data pipeline and Tableau dashboard that turns 11 seasons of race data into a repeatable, evidence-based Investment Scorecard.

From 14 raw CSV files to a clean analytical dataset

Ergast F1 Historical Database - 14 raw CSV files (Kaggle).
Coverage: 11 seasons, 20 constructors, 59 drivers, 228 race events, 32 circuits.

4,626 rows × 25 columns - main table. Pit stop table: 8,360 rows. Constructor standings: 2,319 rows.

V6 Turbo Hybrid regulations redefined the competitive landscape. Pre-2014 data is not comparable for modern sponsorship analysis.

- \N null strings converted to NaN across all 14 files
- 10 sparse columns dropped (>40% null): position, time, milliseconds, fastestLap*, rank, dob
- positionOrder used instead of position — always filled, even for DNFs
- Data types standardised: dates → datetime, numeric → float
- 3 features engineered: position_delta, finished (binary), driver_name
- Pit stops >60 seconds flagged as abnormal, excluded from efficiency KPI

DATA DICTIONARY — KEY FIELDS

positionOrder
Final race finishing rank — always populated, including DNFs.

points
Championship points scored per driver per race.

grid
Starting grid position for each race entry.

finished
Binary flag: 1 = car classified as finished, 0 = DNF.

position_delta
Positions gained or lost during the race (grid – positionOrder).

constructor_name
Team identifier — consistent across the hybrid era.

Engineered fields in red are derived columns not present in the original Ergast export — created specifically for this sponsorship ROI analysis.

KPI framework

Five dimensions, one investment score

Points Momentum (20%)

YoY championship points growth. Identifies rising teams before the market reprices them.

Result Volatility (25%)

Std. deviation of finish positions. Low volatility = predictable brand exposure.

Reliability (15%)

% of races completed. Every DNF means zero broadcast coverage that weekend.

Race Craft (10%)

Avg. positions gained per race. Overtakes and strategy calls amplify brand visibility.

Pit Stop Efficiency

Avg. stop duration (normal stops only). Faster crews signal elite operational discipline.

Investment Score

0.30 Points + 0.25 Consistency + 0.20 Momentum + 0.15 Reliability + 0.10 Race Craft

Key EDA insights

2014–2020
Mercedes dominance is structurally over

25.0% share
Top team's 2024 points share — lowest in the Hybrid Era

0 DNFs · +129%
McLaren: peak reliability and peak growth in 2024

-32% YoY
Red Bull's 2024 points drop — regulatory cycle, not structure

~5 sec gap
Best vs. worst pit crew — multiple grid positions per race

DNFs down since 2019
F1 delivers more guaranteed sponsor exposure per race weekend

Based on exploratory analysis across 11 seasons of Hybrid Era F1 data (2014–2024)

Advanced analysis

Key correlation: avg_finish_position ↔ total points $r = -0.92$ — strongest predictor in the dataset

Volatility vs season points

$R^2 = 0.28$ · $p < 0.001$

Lower result volatility explains 28% of season points variance. Consistency is a validated investment signal, not just a preference.

Pit stop speed vs championship points

Cohen's $d = 1.26$ · $p < 0.001$ (large effect)

Teams with faster-than-median pit stops score significantly more points. Pit stop performance belongs in every sponsor's due diligence checklist.

Drivers of season points

$\beta = -140.78$ for avg_finish_position

Every 1-position improvement in average finish = ~140 additional championship points. Average finishing position is the dominant predictor.

Momentum trajectories

McLaren +40.04 pts/yr ($p = 0.011$) · Mercedes −29.84 pts/yr ($p = 0.001$)

Red Bull's positive long-run slope reversed sharply in 2024 — rolling momentum hit −93.5 pts/yr. Trajectory matters as much as current standing.

Dashboard overview

Investment & Risk

- **D1 - Investment Scorecard:** ROI scatter, tier rankings, KPI cards
- McLaren: Gold tier (63.02); Alpine: lowest (29.3)
- **D2 - Portfolio Matrix:** Risk-return frontier, momentum trends
- Portfolio Sharpe Ratio: 41.49; best risk-adjusted: Sauber

Operations & Filters

- **D3 Operational Excellence:** Pit efficiency, DNF waterfall, reliability
- Fastest avg pit: 23.07s; biggest gap: Mercedes vs Lotus F1
- **Filters:** Constructor, season slider, investment tier
- Cross-dashboard linking , team selection highlights across all views

Top insights

Four decisions every F1 sponsor needs to make now

01

Highest-value sponsorship opportunity on the grid.

Investment Score 63.02 - the only Gold tier team. Points growth +128.95% YoY. Zero DNFs across 24 races. Momentum slope +229.5 pts/yr.

Sponsors who act before a second Constructors' title will pay far less than those waiting for market confirmation.

02

The most dangerous sponsorship trap of the current cycle.

2024 momentum: −93.5 pts/yr. Points down 32% YoY.

Sponsors paying 2022–23 premium rates are buying historical prestige, not future exposure.

03

Most undervalued risk-adjusted investment on the grid.

Consistency score: 100. Result volatility 2024: 2.54 (lowest on grid). Investment Score 52.00 at a fraction of McLaren's cost.

Ideal for conservative brands - banks, insurance, pharma - that need guaranteed visibility over flashy wins.

04

The most financially democratic grid since 2014.

Top team points share: 25.05% - lowest in 11 years.

Five teams (Aston Martin, Haas, Williams, Alfa Romeo, Ferrari) show positive growth momentum at below-average sponsorship cost.

Recom-
mendations

Prioritise McLaren partnerships

McLaren's 21 podiums vs Red Bull's 18 in 2024 translates to an estimated $1.5–4.5M additional earned media value per season for primary livery sponsors.
Priority: High | Feasibility: High

Build a diversified portfolio

Allocate budget 60:25:15 across McLaren, Sauber and Aston Martin - above average return, below average volatility, reducing single-team risk from 100% to ~33%.
Priority: High | Feasibility: High

Add performance-linked clauses

Red Bull sponsors received ~32% less exposure value in 2024 than contracted. Performance-based fee adjustments of 15–25% could save a mid-size sponsor $500K–$2M per season.
Priority: High | Feasibility: Medium

Three moves, measurable returns

Cost of inaction: Every season of delay on McLaren narrows the pricing advantage. Sponsors who act after championship confirmation historically pay a 30–50% premium on pre-confirmation rates.

$1.5–4.5M

McLaren premium partnership
3 extra podiums = ~4.5 hrs additional brand exposure.
Priority: High | Feasibility: High

−67%

Diversified portfolio (McLaren + Sauber + Aston Martin)
Efficient Frontier: above-avg return, below-avg volatility.
Priority: High | Feasibility: High

$500K–2M

Performance-linked contract clauses
Red Bull 2024: −32% points vs 2023. Auto fee adjustment of 15–25% could protect exposure value.
Priority: High | Feasibility: Medium

Six honest boundaries of this analysis

1. No financial data. The Investment Scorecard relies entirely on performance proxies - actual sponsorship fees are proprietary. The framework ranks relative investment quality, not absolute ROI.

2. Team continuity gaps. Rebranded teams (Force India → Aston Martin; Toro Rosso → RB F1) are treated as separate entities, understating Red Bull's junior programme continuity and Aston Martin's lineage.

3. COVID-19 anomaly (2020). The 2020 season ran 30% shorter than standard. YoY calculations around 2020–2021 may be distorted by this compressed baseline.

4. Driver vs constructor conflation. Constructor-level analysis cannot isolate individual driver talent - such as the Verstappen effect at Red Bull - from team infrastructure quality.

5. Data recency. The dataset covers up to end of 2024. New regulations, driver transfers, and technical changes from 2025 onwards are not captured.

6. Pit stop sample sizes. Smaller teams with earlier exits have limited pit stop records in specific seasons, producing potentially unreliable average duration values for those data points.

Next steps

Five extensions to strengthen the investment framework

Predictive modelling

Apply ARIMA or Prophet forecasting to each constructor's points trajectory to predict the **2025 championship ranking** — turning the scorecard forward-looking.

Driver-adjusted ratings

Compute a **driver premium score** to separate team infrastructure quality from individual talent — producing a purer investment metric.

Real-time integration

Connect to the **F1 live timing API** for an in-season monitoring dashboard that updates after every race weekend.

Sponsor ROI linkage

Link **brand awareness surveys and social sentiment** to race results to build a true off-track ROI model beyond performance proxies.

Circuit-weighted scoring

Weight points at **high-viewership circuits** (Monaco, Singapore, Monza) more heavily to reflect disproportionate global media value per race.

F1 is a **$2.6B sponsorship market** still driven by gut feel. This project delivers a five-dimension, statistically validated Investment Scorecard — giving sponsors a repeatable, data-driven alternative. **McLaren is the clear entry point.** The methodology is replicable for any future season.